# Maximilian Rücker

Ulm University  maximilian-ruecker@web.de  Maximilian Rücker  RueckerM

## Education

**Ulm University** May 2023 – today
*Ph.D. Mathematics*
- Research topic: High-dimensional panel data models.
- Supervisor: Prof. Dr. Michael Vogt

**Ulm University** Mar 2021 – Mar 2023
*M.Sc. Mathematics*
- Thesis: "High-Dimensional Inference with the Lasso."
- Final Grade: 1.2

**Ulm University** Oct 2017 – Mar 2021
*B.Sc. Business Mathematics*
- Thesis: "Fractional Poisson Processes and Fractional Poisson Fields."
- Final Grade: 1.8

*German grading scale: 1.0 is the highest grade.*

## Publications

**Estimation and Inference in High-Dimensional Panel Data Models with Interactive Fixed Effects** Nov 2025
*Quantitative Economics*, 2025.
Joint with Michael Vogt, Oliver Linton and Christopher Walsh.

## Job Market Paper

**High-Dimensional Panel Data Models with Interactive Fixed Effects: Beyond the Linear Case** Aug 2026
*arXiv preprint*, `http://arxiv.org/abs/2608.02055`.
Joint with Michael Vogt and Oliver Linton.

**Abstract.** Modern economic panel data sets are often high-dimensional: they contain information on a wide variety of control variables whose number may even exceed the sample size. Nevertheless, the literature on econometric methods for high-dimensional panels is quite limited. In this paper, we study high-dimensional panel models with interactive fixed effects where the regression function has an additive structure, i.e., each covariate enters the model via an unknown nonlinear component function. We develop estimation methodology and theory in this additive framework which substantially extends previous work on the high-dimensional linear case by Rücker et al. (2025). In the theoretical part of the paper, we derive the convergence rate of our estimator for both the small-$T$ and the large-$T$ panel case. The theory is complemented by comprehensive Monte Carlo experiments and an empirical application.

## Work in Progress

**Inference in high-dimensional additive models**
Joint with Michael Vogt.

**Abstract.** In this work, we develop a specification test for a nonlinear treatment effect in the high-dimensional additive model. We establish asymptotic validity under the null hypothesis "no treatment effect" and consistency under fixed alternatives satisfying a standard completeness condition.

## Selected Talks

**31st International Panel Data Conference** July 2026
Speaker at the IPDC 2026 in Exeter about "High-Dimensional Panel Data Models with Interactive Fixed Effects: Beyond the Linear Case".

**Cambridge Econometrics Seminar (Ph.D.)** Mar 2025
Talk about "Additive High-Dimensional Panel Data Models with Interactive Fixed Effects."

**29th International Panel Data Conference** July 2024
Speaker at the IPDC 2024 in Orléans about "Estimation and Inference in High-Dimensional Panel Data Models with Interactive Fixed Effects".

## Scientific Activities

**H2O Workshop Aarhus** Mar 2026
Participant and poster presentation at the H2O (Higher-Order pattern-discovery in High-dimensional data) conference in Aarhus.

**Workshop organisation** July 2025
Organisation of a workshop for PhD students in Mathematics at Ulm University.

**University of Cambridge** Mar 2025
One month research stay at the University of Cambridge supervised by Oliver Linton

## Statistical Packages

**R-package** `hdcce` Nov 2025
Available at CRAN for the HD-CCE–estimator proposed in Rücker et al. (2025).

**R-package** `LassoNoiseInference` Mar 2023
Available at GitHub for the estimation of the Lasso's effective noise proposed in "Estimating the Lasso's Effective Noise" (2021) by Johannes Lederer and Michael Vogt.

## Teaching Experience

| | |
|---|---|
| **Teaching assistant** | Oct 2024 – today |
| Nonparametric statistics (Bachelor's level) | Winter term 25 |
| Asymptotic statistics (Master's level) | Summer term 25 |
| High-dimensional statistics (Master's level) | Winter term 24 |
| **Scientific assistant** | May 2023 – Sept 2024 |
| **Student assistant** | Oct 2019 – Mar 2023 |
| Calculus II (Bachelor's level) | Winter term 22 |
| Calculus I (Bachelor's level) | Summer term 22 |
| Econometrics (Bachelor's level) | Winter term 21 |
| Stochastic processes and Probability theory (Bachelor's level) | Summer term 21 |
| Stochastic processes and Probability theory (Bachelor's level) | Summer term 20 |
| Measure theory (Bachelor's level) | Winter term 19 |

## Refereeing

Journal of Econometrics.

## Software

**Programming languages**
R, Python, Java and MATLAB.
**Technologies**
LaTeX, GitHub, HTML and CSS.

## References

**Michael Vogt**
Institute of Statistics
Ulm University
✉ `m.vogt[at]uni-ulm.de`

**Oliver Linton**
Faculty of Economics
University of Cambridge
✉ `obl20[at]cam.ac.uk`

**Jan Beyersmann**
Institute of Statistics
Ulm University
✉ `jan.beyersmann[at]uni-ulm.de`